

Scattered Spring: How Temperature Synchronizes Ecosystems and its Consequences for Climate Change

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Abstract

A wide variety of biological processes are triggered by cumulative temperatures. For example, the law of flowering plants states that a bud blooms once the temperature sum exceeds a threshold. Similar temperature-triggered mechanisms govern many other aspects of plant development, as well as insect emergence, hibernation, and migration. In this way, cumulative temperatures coordinate the timing of ecological events within an ecosystem. But how reliable is temperature-based coordination—and is that reliability impacted by climate change?

To answer these questions, we model temperature-triggered biological processes as stopped random walks. Cumulative temperature determines the position of the walk, and the occurrence of the biological event is the stopping time. This framework yields two theory-driven implications for how warming alters the timing and synchrony of seasonal events. First, the relationship between temperature and the expected event time is nonlinear. As temperatures increase, the average springtime event such as flowering occurs earlier, but the rate at which they advance can diminish or even accelerate. Second, warming affects not only the expected time but its variability. As temperatures increase and springtime events shift from the equinox toward the solstice, springtime events become less coordinated, producing a more variable or “scattered” spring. We verify both implications empirically.

Introduction

Many biological events are triggered not by the temperature on a single day, but by the temperature accumulated over many days. A canonical example is the law of the flowering plants, which states that a bud blooms in the spring once cumulative temperatures exceed a plant-specific threshold. This and similar mechanisms synchronize a wide variety of seasonal events in ecosystems—for instance, ensuring that pollinators emerge when flowers are in bloom. They also support an array of forecasting tools in which scientists track cumulative temperature or “growing degree days” to predict harvest dates, manage pest emergence, and assess how shifting temperature patterns reshape ecological communities with climate change.

Shifts to earlier biological events—earlier phenology—had been the clearest ‘fingerprint’ of anthropogenic climate change across both natural and managed systems, but have recently become less predictable (CITES). In recent decades phenological advances have decelerated, advancing less with each degree of warming (CITES). With this, variance in the timing of phenological events has variable declined or increased, without a clear model to predict these shifts (CITES). At the same time, previously unknown or rare phenological events appear more common (CITES). While research has provided myriad explanations for these events—often with directly contrasting predictions and logic (CITES)—no work has examined predictions from the existing cumulative sum model fully.

Scientists have used a basic cumulative sum model for over two centuries, but have rarely if ever integrated the relevant mathematical tools developed in probability and statistics over past decades relevant to this model—specifically renewal theory designed for analyzing stopped random walks. Despite the shared mathematical structure, few have connected growing degree-day models and the modern limit theory of stopped random walks.

Here we draw on renewal theory to predict how events triggered by a cumulative temperature sum will shift with warming. Our model predicts observed advances as well as shifts in variance. We show predictions depend critically on characteristics of the underlying temperature regime—variance in the timing of biological events can decrease or increase with warming depending on whether temperatures during the period of accumulation are stationary (as in the winter and summer) or increasing (as in the spring). Taken together our results show how climate change can (i) move spring events earlier at an increasing rate and (ii) increase the variance of their timings, producing less synchronized events as they arrive earlier.

Results & Discussion

Spring events, such as plant flowering, occur on the date on which the cumulative sum first crosses a plant-specific temperature threshold. This places biological timing problems within the domain of stochastic process theory, allowing us to asymptotically characterize the distribution of event timings (hitting times) and thus predict temperature-triggered biological events.

We model effective temperatures (X_i)—the temperatures experienced by a plant bud at a given location—as a baseline temperature (α) with a within-season warming rate ($\{\epsilon_i\}_{i \geq 1}$) commonly seen in the spring months (Fig 1) and some noise (ϵ_i , thus $X_i = \alpha + \beta i + \epsilon_i$). Noise in this formulation could represent many biological aspects depending on the scale and our developing understanding of what triggers spring plant events; for example, in our focus on the bud-level it represents idiosyncratic variation at that bud, which could be driven potentially by microclimate variation (CITES), including bud color variation (CITES), or internal differences that vary when the bud starts accumulating temperatures (CITE molecular chill papers). Bloom occurs on the day (n) the cumulative temperature ($Z_n = \sum_{i=1}^n X_i$ where $t > 0$ is a plant-specific temperature-sum

threshold; $\nu(t) = \min\{m : Z_m > t\}$). Thus $\nu(t)$ is the bloom date—and the hitting time at which cumulative temperatures first exceed the threshold. Standard results for stopped random walks yield asymptotic normal approximations for $\nu(t)$, with behavior governed by the parameters α and β .

A central implication of this framework is that predictions depend strongly on the underlying temperature regime during which the plant accumulates temperature. When plants accumulate temperatures during times when temperature is relatively flat (stationary, common in winter months following the winter solstice) cumulative temperature grows at an approximately linear rate; thus the expected bloom date advances with climate change in step with increasing baseline temperatures ($\mathbb{E}[\nu(t)] \approx \frac{t}{\alpha}$, for example) as is often expected in the literature (CITES). In contrast, plants may accumulate temperatures mainly during the spring months when average daily temperatures increase approximately linearly over time and thus cumulative temperatures grow quadratically. In this regime, bloom date advances as a function of the rate of within-season warming ($\mathbb{E}[\nu(t)] \approx \sqrt{\frac{2t}{\beta}}$)—that is, how quickly spring temperatures increase.

In both regimes, increasing temperatures advance bloom dates in a non-linear function, similar to shifts in a suite of recent papers (CITES). Based on either regime, increasing temperatures advance bloom dates with diminishing advances as the rate of warming (to the baseline temperature, α or within-season rate, β) increases relative to the size of the total threshold (for the stationary regime: $\frac{\partial}{\partial \alpha} \mathbb{E}[\nu(t)] \approx -\frac{t}{\alpha^2}$; for the within-season increasing temperatures regime: $\frac{\partial}{\partial \beta} \mathbb{E}[\nu(t)] \approx -\frac{1}{2}\sqrt{\frac{2t}{\beta^3}}$). Additionally, if climate change shifts the period when plants accumulate temperatures from the non-stationary springtime period (near the spring equinox) to the more stationary period earlier in the year (near the winter solstice) then advances may actually increase, particularly if increases in the baseline (α) outpace increases in the within-season rate (β).

Importantly, these two temperature regimes make very different predictions for the synchrony of events—that is the variance in event timings within the same year and location. Both predict decreased variance with warming (stationary regime: $\text{Var}(\nu(t)) \approx \frac{\sigma^2 t}{\alpha^3}$; non-stationary regime: $\text{Var}(\nu(t)) \approx \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}$), but the influence of noise is strikingly different. In the non-stationary regime—common in the spring—noise is quickly overwhelmed by the effect of increasing within-season temperatures. This effectively acts to synchronize plants (and buds within plants) with the same thermal threshold (aka GDD) requirement—making small difference in microclimate or other idiosyncratic effects negligible. In contrast, under the stationary regime—common in winter—noise can remain relatively influential so that small microclimate differences translate into larger differences in threshold-crossing times. This prediction fits well with experiments conducted under constant temperature regimes, which find increasing variance with decreasing baseline temperatures (CITES). In such experiments, buds often have extremely variable budburst timings, exactly as our model predicts (see Appendix).

These contrasting predictions for variance based on the underlying temperature regime predict increasing variance when anthropogenic warming produces regime shifts. As warming advances events earlier in the calendar, thresholds are reached in portions of the season with flatter mean temperatures where timing can

be more sensitive to local variation. In such cases, warming produces a regime shift that simultaneously (i) advances springtime events potentially at a faster rate and (ii) reduces synchrony across nearby locations, producing a more variable and less coordinated onset of spring even as it arrives sooner.

Such a regime-shift would also cause major shifts in the variance. ...

A central implication of this framework is that synchrony depends on how quickly temperatures are changing at the time the threshold is reached. For example, many temperate climates exhibit two different regimes when flowers first bloom in the spring. In the first regime—roughly the winter months following the winter solstice (January–February)—average daily temperatures are approximately constant over time, so cumulative temperature grows at an approximately linear rate. In the second regime—roughly the spring months around the equinox (March–April)—average daily temperatures increase approximately linearly over time, so cumulative temperatures grows quadratically.

To see how, note that the cumulative temperature can be divided into the cumulative signal (the sum of the averages) plus the cumulative noise (the sum of the random deviations). When the distribution of the noise is roughly the same in both regimes, the quadratic signal in regime 2 can much more quickly overwhelm the noise when compared to the linear signal in regime 1. For this reason, nearby buds tend to cross the threshold within a narrow time window when events occur in regime 2. In contrast, when thresholds are reached in the regime 1, cumulative noise can remain relatively influential so that small microclimate differences translate into larger differences in threshold-crossing times.

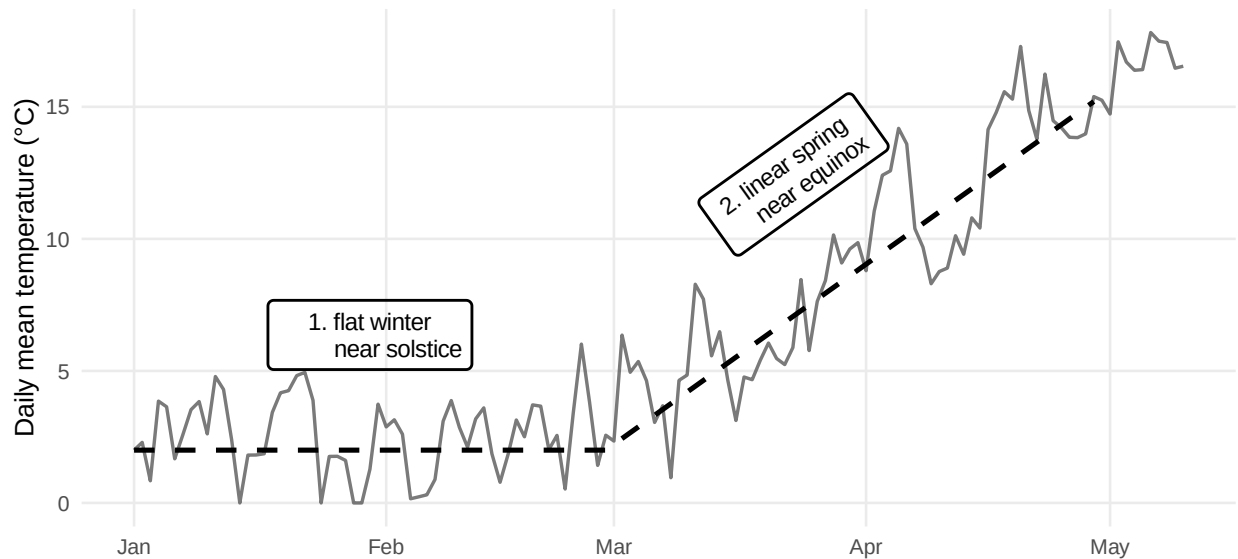


Figure 1: The timing of springtime events in temperate climates depends on whether temperature is accumulated during one of two regimes, 1. a flat winter or 2. a linear spring.

Warming advances threshold-triggered events because the cumulative total reaches the target sooner. Typically, one would expect the rate of advancement to diminish as temperatures warm. Indeed, when traveling a fixed

distance, one has to double the speed to half the travel time so that the relationship between speed and time is a hyperbola.

The twist is that warming can change which aspects of the temperature drives the timing of threshold-triggered events. If average temperatures increase day over day as it does for springtime events in temperate climates near the equinox, the temperature rate not the baseline level of temperatures largely determined the bloom date. In contrast, if average temperatures are relatively flat day over day as they are in temperate climates closer to the solstice, the baseline temperature determines the boom date. Thus, as warming moves springtime events closer to the solstice, the rate at which their timing occurs can increase, accelerating the advancement of spring

The second finding concerns spread rather than the average. Even if warming makes the mean bloom date earlier, it does not necessarily make timing more predictable. The date depends on both the average temperature trend and idiosyncratic variation induced in part by microclimate factors such as shade, wind, soil moisture, urban heat, and other factors that vary even within a single site. When average temperatures rise quickly and steadily, as when closer to the equinox, those local fluctuations are relatively small compared with the strong average temperature “signal,” so nearby buds tend to cross the threshold within a narrow window. When average temperatures are relatively constant, as when closer to the solstice, the seasonal signal is weaker, and small local differences can have outsized effects. For example, one bud may cross after an early warm spell while another must wait for the next, widening the range of dates.

Therefore, warming can both increase or decrease advancement and variability depending on where in the seasonal cycle the threshold is reached. In particular, if warming advances an event into an earlier period with weaker accumulation, microclimate noise has more influence, and synchrony can break down. This regime shift produces a “scattered spring” when climate change moves events earlier on average while simultaneously reducing coordination across individuals and locations, making spring less tightly synchronized even as it arrives sooner.

2. Methodology

To fix ideas, consider the day a bud blooms at a given location. We model the effective temperature experienced by that bud on day i by a linear trend plus microclimate noise,

$$X_i = \alpha + \beta i + \epsilon_i, \tag{1}$$

where α is a baseline level, β is a within-season warming rate, and $\{\epsilon_i\}_{i \geq 1}$ are independent and identically distributed with $\mathbb{E}[\epsilon_i] = 0$ and $\text{Var}(\epsilon_i) = \sigma^2$, representing idiosyncratic variation at that bud such as due to the day-to-day microclimate.

Let

$$Z_n = \sum_{i=1}^n X_i \quad \text{and} \quad \nu(t) = \min\{m : Z_m > t\} \quad (2)$$

denote cumulative temperature by day n and the observed bloom date, respectively, where $t > 0$ is a plant-specific temperature-sum threshold. Thus $\nu(t)$ is the hitting time at which cumulative temperatures first exceed the threshold.

Standard results for stopped random walks yield asymptotic normal approximations for $\nu(t)$, with behavior governed by the parameters α and β . We consider two cases.

Regime 1: constant drift ($\beta = 0$) When the drift is approximately constant and positive,

$$\nu(t) \sim \text{Normal}\left(\frac{t}{\alpha}, \frac{\sigma^2 t}{\alpha^3}\right), \quad (3)$$

see Gut (2011) and Gut (2009, chap. 3).

Regime 2: increasing drift ($\beta > 0$) When average temperatures increase, the hitting time satisfies

$$\nu(t) \sim \text{Normal}\left(\sqrt{\frac{2t}{\beta}} - \frac{\alpha}{\beta} + \frac{1}{2}, \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right), \quad (4)$$

with the derivation sketched in Appendix A.1 and simulations confirming the derivations in Appendix A.3. The functional central limit theorem and linearization argument underlying these results hold somewhat generally, and thus similar limits can be derived when the $\{\epsilon_i\}_{i \geq 1}$ are weakly correlated or the temperature trajectory is not linear but sufficiently.

From equations (3)–(4), we make two observations. Finding 1: there is a nonlinear relationship between warming and the mean timing of springtime events, and warming can advance springtime events at an increasing rate. Finding 2: the two regimes are not equally sensitive to the idiosyncratic variation, and thus warming can increase the variation of springtime events, thereby scattering spring. We now discuss both findings in greater detail.

2.1 Finding 1: warming can advance springtime events at an increasing rate

In both regimes, the expected bloom date is a nonlinear function of the warming parameters.

Regime 1: constant drift ($\beta = 0$) From (3),

$$\mathbb{E}[\nu(t)] \approx \frac{t}{\alpha}, \quad \text{Var}(\nu(t)) \approx \frac{\sigma^2 t}{\alpha^3}. \quad (5)$$

Increasing baseline temperatures (larger α) advances the expected bloom date, but the marginal advance diminishes since

$$\frac{\partial}{\partial \alpha} \mathbb{E}[\nu(t)] \approx -\frac{t}{\alpha^2}, \quad (6)$$

where the magnitude decreases as α grows.

Regime 2: increasing drift ($\beta > 0$) From (4), when t is large,

$$\mathbb{E}[\nu(t)] \approx \sqrt{\frac{2t}{\beta}}, \quad \text{Var}(\nu(t)) \approx \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}. \quad (7)$$

Increasing the within-regime warming rate (larger β) also advances the expected bloom date with diminishing returns since

$$\frac{\partial}{\partial \beta} \mathbb{E}[\nu(t)] \approx -\frac{1}{2} \sqrt{\frac{2t}{\beta^3}}. \quad (8)$$

However, if warming climates shift the bloom date from regime 2 to regime 1, then the rate of advancement can increase, particularly if increases in α outpace increases in β .

2.2 Finding 2: warming can scatter spring events

Equations (3)–(4) also clarify how warming affects dispersion. Holding the regime fixed, warming reduces variance. That is, $\text{Var}(\nu(t))$ decreases with α in (3) and with β in (4). However, just as in Section 2.1, warming can change which regime is relevant at the realized bloom date. When an event is advanced sufficiently early, microclimate fluctuations play a larger role in determining the hitting time.

One way to quantify sensitivity to microclimate variation is through the (squared) coefficient of variation,

$$\text{CV}^2 = \frac{\text{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2}.$$

In regime 1, $\mathbb{E}[\nu(t)] \approx t/\alpha$ and $\text{Var}(\nu(t)) \approx \sigma^2 t/\alpha^3$, so

$$\text{CV}_1^2 = \frac{\text{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2} \approx \frac{\sigma^2 t/\alpha^3}{(t/\alpha)^2} = \frac{\sigma^2}{\alpha t}.$$

In regime 2, $\mathbb{E}[\nu(t)] \approx \sqrt{2t/\beta}$ and $\text{Var}(\nu(t)) \approx \sigma^2/(\beta^{3/2}\sqrt{2t})$, hence

$$\text{CV}_2^2 = \frac{\text{Var}(\nu(t))}{\mathbb{E}[\nu(t)]^2} \approx \frac{\sigma^2/(\beta^{3/2}\sqrt{2t})}{(2t/\beta)} = \frac{\sigma^2}{\sqrt{\beta} (2t)^{3/2}}.$$

Both expressions increase when deterministic accumulation is weaker relative to the microclimate scale σ (e.g., smaller α and/or smaller effective β). Consequently, warming tends to reduce variability within a given seasonal regime by strengthening deterministic forcing.

To compare regimes, note that

$$\frac{CV_1^2}{CV_2^2} \approx \frac{\sigma^2/(\alpha t)}{\sigma^2/(\sqrt{\beta}(2t)^{3/2})} = \frac{\sqrt{\beta} 2^{3/2}}{\alpha} \sqrt{t}.$$

Thus, since α and β are nearly always greater than 0, the regime 1 becomes arbitrarily more sensitive to microclimate fluctuations than regime 1 for large t . This produces a more variable, less synchronized and thus “scattered” spring when warming shifts threshold crossings from regime 2 to regime 1.

3. Application

We apply the two models described in Section 2 to two datasets. The first is dataset is from an experiment conducted by Charrier et al. (2011), which demonstrates regime 1. The second is a large dataset of nearly 24,000 observations from Rosemartin et al. (2015), which demonstrate the shift between regimes. Unfortunately, we are unable to find an experiment that demonstrates regime 2 or the shift between regimes.

3.2 Observational Evidence

We now consider data from the historical lilac–honeysuckle phenology network compiled by Rosemartin et al. (2015), which contains leafing and flowering dates collected across the continental United States from 1956–2014 for purple common lilac (*Syringa vulgaris*), a cloned lilac cultivar (*S.*, $\times$, *chinensis* ‘Red Rothomagensis’), and two cloned honeysuckle cultivars (*Lonicera tatarica* Arnold Red and *L. korolkowii* Zabeli). We focus site that recorded the phenophase full bloom and are within 10 miles of a GHCND site with complete temperature records, resulting in 23,864 observations.

For each observation, we estimate α using the average daily temperature between January and February. We estimate β using the slope of a linear regression model fit by regressing daily temperature on day index over March and April. Temperatures below 0 are set to 0. The parameters α and β are shown for nine random sites in Figures 4 and 5 in Appendix A.2, which have the same interpretation as Figure 1. Site 14726 in New Mexico (Year 1969) is a good example of a bloom in regime 2 since relatively little temperature was accumulated during January and February. The average bloom date was May 8th with a standard deviation of 17 days. Site 14276 in New York (Year 1982) is a good example of a bloom between regimes 1 and 2, which had an average bloom date of March 18 with a standard deviation of 27.

The α and β typically increase over time as climates warm as demonstrated by Figures 3 and 4 in the Appendix A.2. Each figure shows α and β (points) for the nine sites in which observations were made for the most number of years. The solid lines are regression lines and the grey intervals are 95 percent confidence bands.

To confirm the predictions from Section 2, we first bin sites and years within a grid of α (rows) and β

Table 1: Mean bloom date (day-of-year) by average Jan-Feb temperature α (rows) and average Mar-Apr temperature increase β (columns).

$\alpha \setminus \beta$	$[-0.28, 0.07]$	$(0.07, 0.12]$	$(0.12, 0.16]$	$(0.16, 0.68]$
$(0, 0.6]$	163.05	154.24	148.96	142.56
$(0.6, 1.6]$	153.21	147.97	142.60	136.78
$(1.6, 4]$	138.86	133.56	130.06	126.34
$(4, 16.4]$	105.13	106.46	105.26	102.67

Table 2: Standard deviation of bloom date (days) by winter level α (rows) and spring warming rate β (columns).

$\alpha \setminus \beta$	$[-0.28, 0.07]$	$(0.07, 0.12]$	$(0.12, 0.16]$	$(0.16, 0.68]$
$(0, 0.6]$	16.08	12.36	10.44	9.61
$(0.6, 1.6]$	15.24	12.87	13.20	12.67
$(1.6, 4]$	17.97	16.52	16.31	14.56
$(4, 16.4]$	22.78	19.84	17.23	14.76

(columns) and calculate the mean and standard deviation of the observed bloom date (number of days since January 1) within each bin. Table 1 shows the mean bloom dates by (α, β) bin, chosen from the α and β quartiles. Note that although the model in Section 2 assumes $\beta \geq 0$, the empirical estimates of β may be near-zero or even negative in anomalous years. We therefore treat the lowest β bin as an approximately no increase category.

Several patterns match the predictions of the model from Section 2. First, holding α fixed, larger β is associated with earlier mean bloom dates, which generally but not always advance at a decreasing rate. The same is true for holding β fixed, and increasing α . Holding α at its largest value, the influence of β is considerably smaller. This reflects the accuracy of regime 1 where β plays no role.

Table 2 reports the standard deviation by bin (chosen based on the quartiles of α and β). Consistent with the theory, dispersion generally declines as β increases within each α row. This occurs because when α is fixed, the percent of buds that bloom in regime 1 is unchanged, and the buds that makes it to regime 2 have a lower variance. However, holding β fixed and increasing α produces a regime shift in which more buds bloom under regimes 1 and thus the bloom date dispersion increases. This is the “scattered spring” predicted in Section 2.

Aggregating by quartiles makes it difficult to determine the rate at which the mean and variance of the bloom times are increasing and can potentially result in aggregation bias. For this reason, we check Tables 1–2 by replacing the binning with a smooth, nonparametric estimate of the conditional mean and dispersion using generalized additive models, see Hastie and Tibshirani (1986) and Wood (2017). Specifically, we fit a two-parameter Gaussian location–scale model using the `gaulss` function from `mgcv`:

$$\mu(\alpha, \beta) = \mathbb{E}[\nu(\tau) \mid \alpha, \beta], \quad \sigma(\alpha, \beta) = \text{SD}(\nu(\tau) \mid \alpha, \beta),$$

where both μ and σ are modeled with tensor-product splines with basis dimension $k = 10$. This approach uses all observations while allowing flexible interactions between the two temperature summaries.

The results are displayed in Figure 2, which show the mean (left) and log standard deviation (right) plotted against α and β . The left figure shows the mean generally decreases at an increasing rate as α and β increase—although increases are possible when β is large. The standard deviation can increase or decrease, again depending on the relative values of α or β .

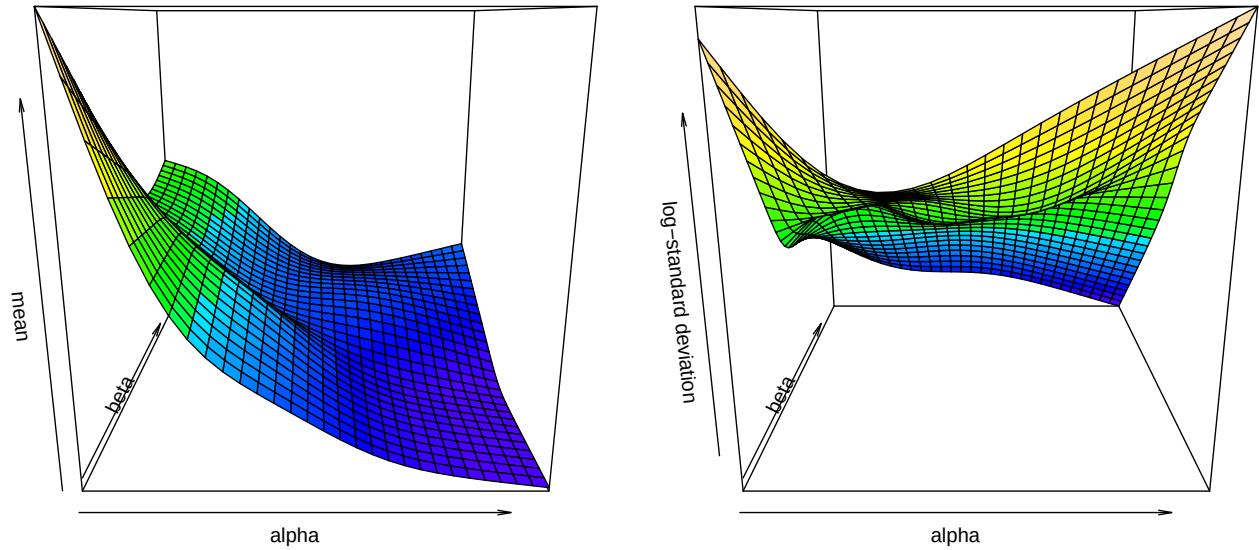


Figure 2: Mean and log standard deviation of lilac bloom dates ($n = 23,864$) by α and β estimated by a generalized additive model.

4. Discussion

In this paper, we show that when biological processes use cumulative temperatures to coordinate the timing of events, that coordination depends on whether average temperatures are increasing or not. As bloom dates advance into regions in which average temperatures do not increase much day today, microclimate variation plays a much more important role, reducing ecosystem synchrony and scattering spring over a longer time period.

The exact effect on an ecosystem depends on how exactly α and β change. In some locations, climate change may manifest primarily as higher winter temperatures (an increase in α) with relatively little change in the rate at which temperatures rise during spring (a modest change in β). In others, winter temperatures may change less while spring transitions steepen (a larger increase in β). Because the mean and variance of the hitting time depend differently on α and β , these distinct warming profiles can produce different phenological responses even when average annual warming is similar.

Variation in α and β levels are shown in Figures 2-3 in Appendix A.2. We chose the nine sites in the lilac

dataset described in Section 3.2 with the longest data collections. In many cases, α has increased but β has stayed the same. The result is the increased variance and decreased synchrony that we documented in Sections 2 and 3.

This heterogeneity matters most for synchrony. Within a fixed seasonal regime (regime 1 or 2), higher α in winter or higher β in spring tends to reduce dispersion. However, increases in α can also shift threshold crossings earlier in the calendar, potentially moving events from regime 2 into regime 1, where microclimate variability has greater influence on timing. Whether warming leads to a more synchronized or more “scattered” spring therefore depends on the local joint evolution of (α, β) and on the threshold t relevant for a given species and life-cycle event. In this sense, the stopped-random-walk framework offers a simple way to translate spatially varying climate change into spatially varying predictions about both the advancement of spring and the stability of ecosystem coordination.

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A. Appendix

A.1 Sketch of Theoretical Results

We follow the notation in Gut (2009, chap. 6). Let $X_i = \alpha + \beta i + \epsilon_i$ denote the temperature on day i , where $\alpha > 0$, $\beta \geq 0$, $\mathbb{E}[\epsilon_i] = 0$, and $\text{Var}(\epsilon_i) = \sigma^2$. Define the deterministic cumulative sum

$$\xi_n = \sum_{i=1}^n \mathbb{E}[X_i] = \alpha n + \frac{\beta}{2}n(n+1) = \frac{\beta}{2}n^2 + \beta\gamma n, \quad \gamma = \frac{\alpha}{\beta} + \frac{1}{2},$$

with $\xi_n = \alpha n$ when $\beta = 0$. Denote the sum of the stochastic component $S_n = \sum_{i=1}^n \epsilon_i$. Our object of interest is the first-passage time of

$$Z_n = S_n + \xi_n, \quad \nu(t) = \min\{n \geq 1 : Z_n > t\}.$$

In the language of Lai and Siegmund (1977), $\{Z_n\}$ is a perturbed random walk: the random walk $\{S_n\}$ is perturbed by a deterministic term $\{\xi_n\}$, and when $\beta > 0$ the quadratic growth of ξ_n dominates long-run behavior. This setup yields two asymptotic regimes, corresponding to the empirical distinction between regime 1 ($\beta \approx 0$) and regime 2 ($\beta > 0$), which in turn underlies our qualitative predictions about nonlinear mean shifts and changes in dispersion.

In regime 1, $\beta = 0$, and we apply the usual asymptotic argument for the hitting time of a random walk with constant positive drift.

$$\nu(t) \sim \text{Normal}\left(\frac{t}{\alpha}, \frac{\sigma^2 t}{\alpha^3}\right),$$

see Gut (2011) and Gut (2009, chap. 3).

In regime 2, $\beta > 0$, and the deterministic component $\xi_n = \frac{\beta}{2}n^2 + \beta\gamma n$ grows quadratically so that the threshold t is reached near the deterministic crossing time $m(t)$ solving $\xi_{m(t)} \approx t$, i.e.

$$m(t) = \frac{-\beta\gamma + \sqrt{\beta^2\gamma^2 + 2\beta t}}{\beta} = \sqrt{\frac{2t}{\beta}} - \gamma + o(1).$$

By the functional central limit theorem, the deviation of $\nu(t)$ about $m(t)$ is approximately normal. Linearization of $Z_{\nu(t)} \approx t$ around $m(t)$ yields

$$\nu(t) \sim \text{Normal}\left(m(t), \frac{\sigma^2 m(t)}{(\alpha + \beta m(t))^2}\right) = \text{Normal}\left(\sqrt{\frac{2t}{\beta}} - \gamma, \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right),$$

where we have used the fact that $\alpha + \beta m(t) \sim \beta m(t) \sim \sqrt{2\beta t}$ for $t \gg 0$. Thus, in regime 2 the mean hitting time scales like $\sqrt{t}$ (rather than linearly in t) and the variance shrinks at rate $t^{-1/2}$, reflecting that increasing forcing makes microclimate fluctuations progressively less influential near the crossing.

A.2 Additional Figures

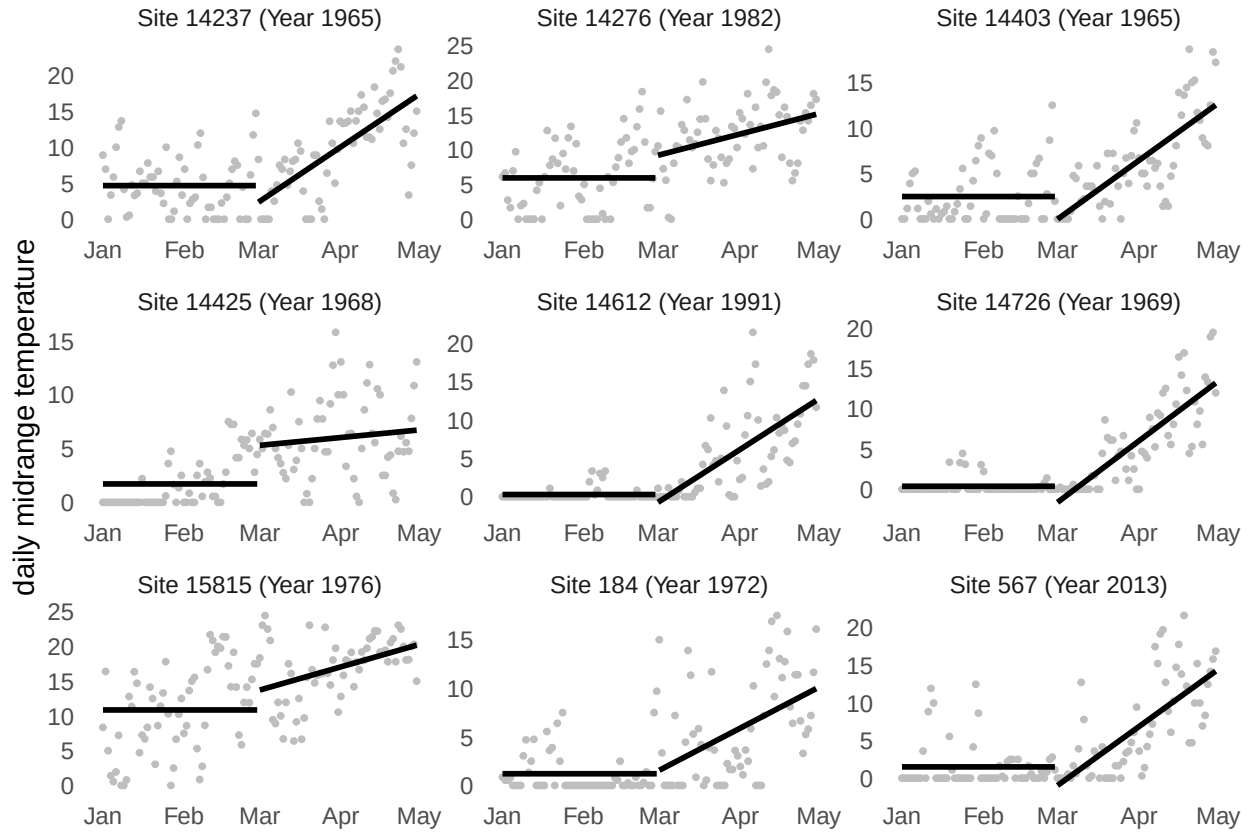


Figure 3: Springtime daily midrange temperatures (points) with α (mean from Jan to Feb) and β (slope from Mar to Apr) for nine randomly selected sites. Temperatures below 0 are set to 0.

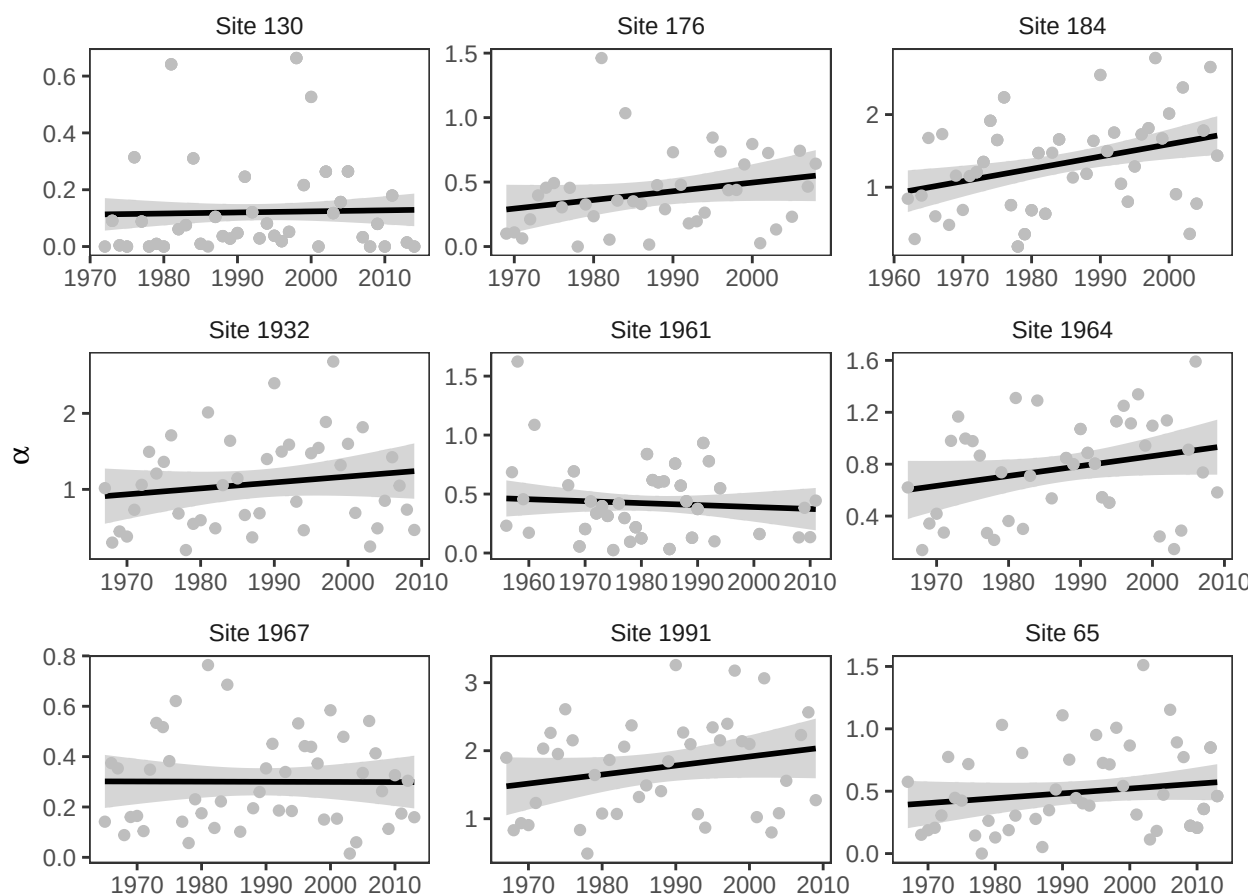


Figure 4: Changes in alpha for 9 sites in lilac data.

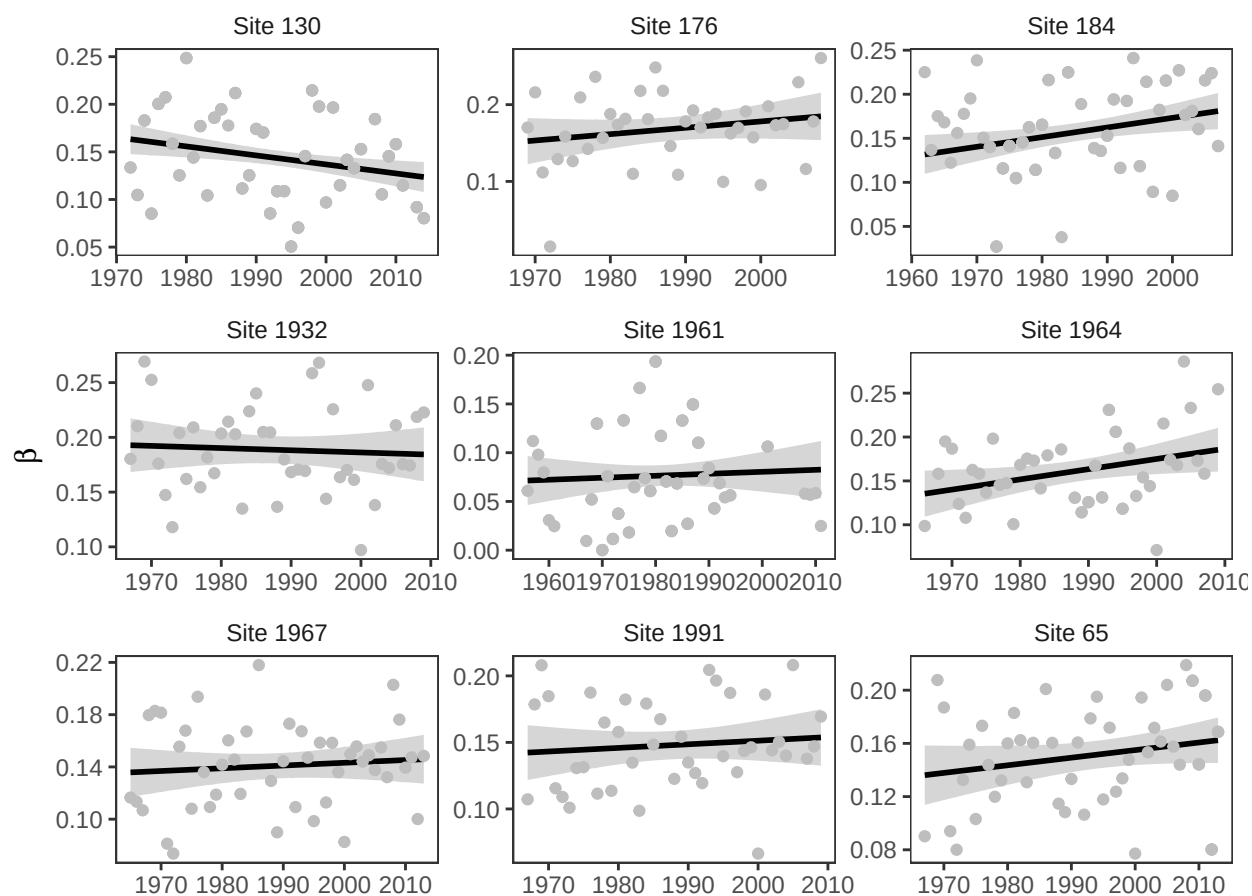


Figure 5: Changes in beta for 9 sites in lilac data.

A.3 Simulations

We run two simulations to study the results from Sections 2 and 3. The first checks the asymptotic approximation from Section 2. The second checks the interpretation of the data from Section 3.

Simulation 1

The first simulation verifies the accuracy of the two asymptotic normal approximations for the hitting time

$$\nu(t) = \min \left\{ n \geq 1 : \sum_{i=1}^n X_i > t \right\}$$

under the model

$$X_i = \alpha + \beta i + \varepsilon_i, \quad \varepsilon_i \sim \text{Normal}(0, \sigma^2).$$

For simulation, we set $\sigma = 20$ and generate daily temperatures until the threshold t is reached, recording the corresponding hitting time $\nu(t)$. We repeat this procedure $R = 10,000$ times for thresholds $t = \{1000, 2000\}$, $\alpha = \{2, 4\}$ and $\beta = \{0, 0.1\}$. When $\beta = 0$ the simulation matches the winter regimes, and when $\beta = 0.1$, it matches the spring regime. The simulated distribution of $\nu(t)$ is represented in Figure 6 below by the histograms.

We then compare these simulations with the theoretical approximations, we standardized each simulated hitting time using the relevant asymptotic mean and standard deviation. When $\beta = 0$, we use the winter regime approximation

$$\nu(t) \sim \text{Normal}\left(\frac{t}{\alpha}, \frac{\sigma^2 t}{\alpha^3}\right).$$

When $\beta > 0$, we use the spring regime approximation

$$\nu(t) \sim \text{Normal}\left(\sqrt{\frac{2t}{\beta}} - \frac{\alpha}{\beta} + \frac{1}{2}, \frac{\sigma^2}{\beta^{3/2}\sqrt{2t}}\right).$$

For each setting we computed the standardized variable

$$z = \frac{\nu(t) - \mu}{\text{sd}},$$

plotted its histogram, and overlaid the standard normal density. Across both regimes, the standardized histograms align closely with the $\text{Normal}(0, 1)$ curve, and the agreement improves as t increases, providing a visual confirmation of the derived asymptotic distributions.

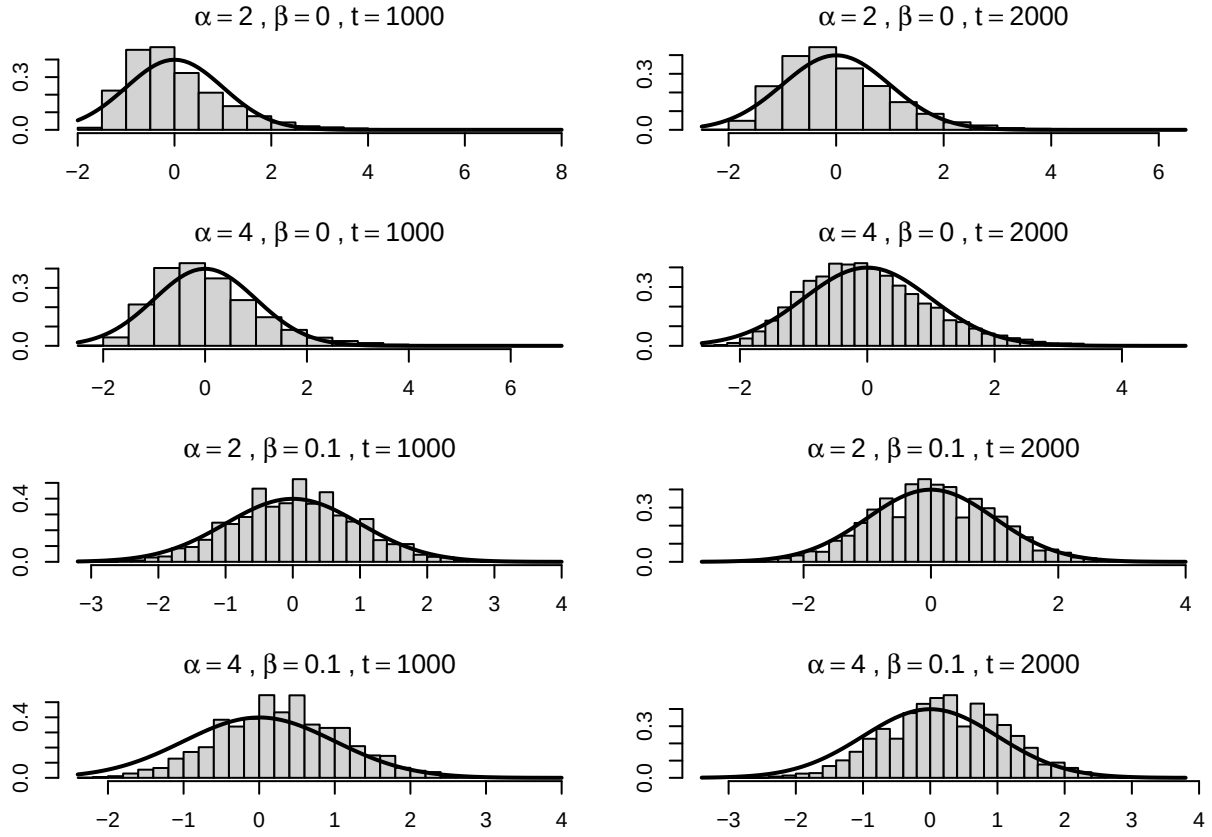


Figure 6: Simulations comparing the distribution of bloom dates under the winter (top) and spring (bottom) regimes.

Simulation 2

To verify our interpretation of the data in Section 3, we simulate the model from section. In each replicate we generated daily temperatures

$$X_i = \mu_i + \varepsilon_i, \quad \varepsilon_i \sim \text{Normal}(0, \sigma^2),$$

with $\sigma = 20$, where the deterministic component follows

$$\mu_i = \begin{cases} \alpha, & i = 1, \dots, 90 \quad (\text{January–February}), \\ \alpha + \beta(i - 90), & i = 91, \dots, 180 \quad (\text{March–April}). \end{cases}$$

Bloom occurs when cumulative temperature first exceeds a threshold t , i.e.

$$\nu(t) = \min \left\{ n \geq 1 : \sum_{i=1}^n X_i > t \right\}.$$

We simulated $R = 10,000$ independent realizations of $\nu(t)$ for each combination of $\alpha \in \{4, 8, 10\}$ and $\beta \in \{0.2, 0.4, 0.8\}$, and for thresholds $t \in \{1000, 2000\}$. For each parameter pair we report the Monte Carlo mean and standard deviation of $\nu(t)$ (Tables~3a–3d), arranged with rows indexing α and columns indexing β .

The results reproduce the qualitative patterns observed in the lilac application. Mean bloom dates decrease as either the winter level α increases or the spring warming rate β increases (Tables~3a and 3c). Within a fixed α row, increasing β reduces the standard deviation of bloom dates (Tables~3b and 3d), consistent with the spring-regime prediction that stronger increasing forcing diminishes the leverage of noise on the threshold-crossing time. Holding β fixed, dispersion is often larger at higher α (particularly for $\beta \in \{0.4, 0.8\}$), consistent with the regime-shift intuition that earlier crossings can occur in a flatter portion of the seasonal cycle where microclimate variability has greater influence.

Table 3: Piecewise winter–spring simulations. Simulated mean and standard deviation of the bloom date $\nu(t)$ across $R = 10,000$ Monte Carlo replicates, for thresholds $t \in \{1000, 2000\}$. Rows index the winter level α and columns index the spring warming rate β .

(a) Mean, $t = 1000$				(c) Mean, $t = 2000$			
$\alpha \setminus \beta$	0.2	0.4	0.8	$\alpha \setminus \beta$	0.2	0.4	0.8
4	151.66	136.83	124.86	4	199.54	171.15	149.16
8	114.87	111.27	106.85	8	169.81	152.43	137.37
10	98.45	97.20	95.72	10	156.22	143.40	131.34

(b) SD, $t = 1000$				(d) SD, $t = 2000$			
$\alpha \setminus \beta$	0.2	0.4	0.8	$\alpha \setminus \beta$	0.2	0.4	0.8
4	15.48	10.42	7.21	4	10.96	7.22	4.84
8	16.53	13.27	10.50	8	10.93	7.50	5.11
10	15.94	14.45	12.71	10	10.69	7.66	5.36